

Step 1 – Understand the Problem and Establish Design Scope

The first step in designing a news feed system is:

Requirement Clarification

System design interview questions are intentionally open-ended.

Therefore, it is important to ask clarification questions before jumping into the design.

The goal is to:

- Understand system requirements
- Identify scale
- Clarify assumptions
- Define scope clearly

Example Candidate - Interviewer Discussion

Platform Support

Candidate:

None

Is this a mobile app, web app, or both?

Interviewer:

None

Both mobile and web applications are supported.

Core Features

Candidate:

None

What are the important features?

Interviewer:

None

Users can publish posts and view friends' posts in the news feed.

Feed Ordering

Candidate:

None

Is the feed ordered chronologically or based on ranking scores?

Interviewer:

None

Assume reverse chronological ordering for simplicity.

This means:

None

Newest posts appear first.

User Relationships

Candidate:

None

How many friends can a user have?

Interviewer:

None

Up to 5000 friends.

Traffic Scale

Candidate:

None

What is the traffic volume?

Interviewer:

None

10 million DAU (Daily Active Users).

Content Type

Candidate:

None

Does the feed contain only text or also media?

Interviewer:

None

Feed can contain images and videos as well.

Functional Requirements

Based on the discussion, the system must support the following functionalities.

1. Post Publishing

Users can:

- Create posts
- Upload images
- Upload videos
- Share updates

2. Feed Retrieval

Users can:

- Open news feed page
- View friends' posts
- View latest updates

Feed Ordering Requirement

The feed is ordered using:

Reverse Chronological Order

Meaning:

None

Most recent posts appear first in the feed.

No complex ranking algorithm is required in this simplified design.

Scale Requirements

The system must support:

Requirement	Value
Daily Active Users	10 million
Maximum Friends per User	5000
Platforms	Mobile + Web
Content Type	Text, Images, Videos

Non-Functional Requirements

Besides core features, the system should also provide:

Requirement	Description
Scalability	Handle millions of users
Low Latency	Fast feed loading
High Availability	Feed accessible at all times
Reliability	No post loss
Fault Tolerance	Continue working during failures

Key Challenges

Designing a large-scale news feed system introduces several challenges:

1. High Read Traffic

Feed retrieval happens very frequently.

2. High Write Traffic

Millions of users publish posts continuously.

3. Media Storage

Images and videos require large-scale storage systems.

4. Feed Fanout

A single post may need to be distributed to thousands of followers.

5. Low Latency

Users expect near real-time feed updates.

Important Assumptions

To simplify the system design:

- Feed ordering uses reverse chronological order
- No advanced recommendation/ranking algorithm
- Friends/followers model is assumed
- Media storage handled separately

Design Focus Areas

After requirement clarification, the next steps usually focus on:

- Feed publishing flow
- Feed retrieval flow
- Fanout strategy

- Cache design
- Database design
- Scalability
- Storage optimization

Key Takeaway

Before designing the architecture, it is critical to:

None

Clearly understand system requirements, traffic scale, and expected features.

This helps create a focused and scalable system design.